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Roll_No:CE11

TASK 1: Import libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

TASK 2: Load Auto MPG dataset

```
url = "https://raw.githubusercontent.com/selva86/datasets/master/Auto.csv"
df = pd.read_csv(url)

print(df.head())
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	year	\
0	18.0	8	307.0	130	3504	12.0	70	
1	15.0	8	350.0	165	3693	11.5	70	
2	18.0	8	318.0	150	3436	11.0	70	
3	16.0	8	304.0	150	3433	12.0	70	
4	17.0	8	302.0	140	3449	10.5	70	

	origin	name
0	1	chevrolet chevelle malibu
1	1	buick skylark 320
2	1	plymouth satellite
3	1	amc rebel sst
4	1	ford torino

TASK 3: Inspect and clean data

```
print(df.info())
print(df.isnull().sum())

df["horsepower"] = pd.to_numeric(
    df["horsepower"],
    errors="coerce"
)

df = df.dropna(
    subset=["horsepower", "mpg"]
)

print("Cleaned data:")
print(df.head())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 392 entries, 0 to 391
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   mpg              392 non-null   float64
1   cylinders        392 non-null   int64
2   displacement     392 non-null   float64
3   horsepower       392 non-null   int64
4   weight           392 non-null   int64
5   acceleration     392 non-null   float64
6   year             392 non-null   int64
7   origin           392 non-null   int64
8   name             392 non-null   object
dtypes: float64(3), int64(5), object(1)
memory usage: 27.7+ KB
None
mpg              0
cylinders         0
displacement     0
horsepower       0
weight           0
acceleration     0
year             0
origin           0
name             0
dtype: int64
Cleaned data:
   mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0  18.0         8         307.0         130    3504          12.0    70
1  15.0         8         350.0         165    3693          11.5    70
```

2	18.0	8	318.0	150	3436	11.0	70
3	16.0	8	304.0	150	3433	12.0	70
4	17.0	8	302.0	140	3449	10.5	70

	origin		name
0	1	chevrolet	chevelle malibu
1	1	buick	skylark 320
2	1	plymouth	satellite
3	1	amc	rebel sst
4	1	ford	torino

```
# TASK 4: Select horsepower and MPG
```

```
X = df["horsepower"].values
y = df["mpg"].values
```

```
print("Horsepower:")
print(X[:10])
```

```
print("MPG:")
print(y[:10])
```

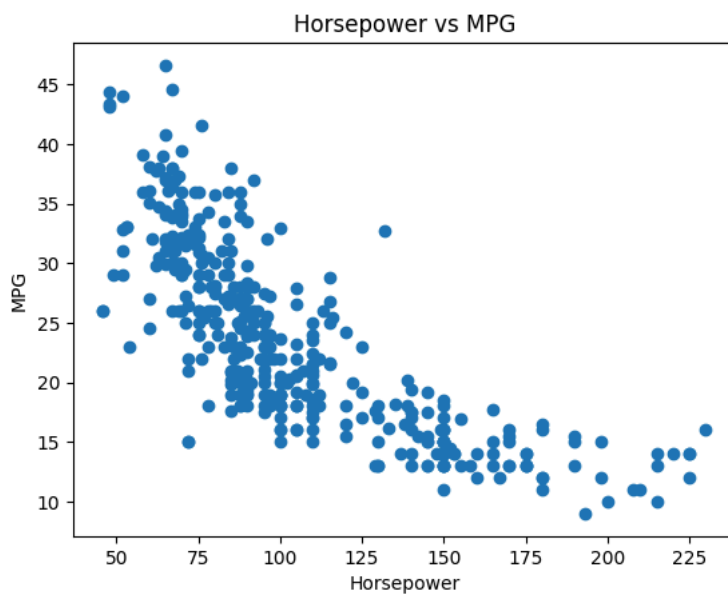
```
Horsepower:
[130 165 150 150 140 198 220 215 225 190]
MPG:
[18. 15. 18. 16. 17. 15. 14. 14. 14. 15.]
```

```
# TASK 5: Plot scatter graph
```

```
plt.scatter(X, y)
```

```
plt.xlabel("Horsepower")
plt.ylabel("MPG")
plt.title("Horsepower vs MPG")
```

```
plt.show()
```



```
# TASK 6: Add bias term
```

```
X = X.reshape(-1, 1)
```

```
X_bias = np.c_[
    np.ones(X.shape[0]),
    X
]
```

```
print("X with bias:")
print(X_bias[:5])
```

```
X with bias:
[[ 1. 130.]
 [ 1. 165.]
 [ 1. 150.]
 [ 1. 150.]
 [ 1. 140.]]
```

```
# TASK 7: Compute correlation

correlation = np.corrcoef(
    df["horsepower"],
    df["mpg"]
)[0, 1]

print("Correlation =", correlation)
```

Correlation = -0.7784267838977756

```
# TASK 8: Calculate OLS weights

# w = (X^T X)^-1 X^T y

w = np.linalg.inv(
    X_bias.T @ X_bias
) @ X_bias.T @ y

print("OLS Weights:")
print("Intercept =", w[0])
print("Slope =", w[1])
```

OLS Weights:
Intercept = 39.9358610211705
Slope = -0.15784473335365387

```
# TASK 9: Predict values

y_pred = X_bias @ w

print("Predicted values:")
print(y_pred[:10])
```

Predicted values:
[19.41604569 13.89148002 16.25915102 16.25915102 17.83759835 8.68260382
 5.21001968 5.99924335 4.42079602 9.94536168]

```
# TASK 10: Compute residuals

residuals = y - y_pred

print("Residuals:")
print(residuals[:10])
```

Residuals:
[-1.41604569 1.10851998 1.74084898 -0.25915102 -0.83759835 6.31739618
 8.78998032 8.00075665 9.57920398 5.05463832]

```
# TASK 11: Calculate SSE, MSE, R²

SSE = np.sum(
    (y - y_pred) ** 2
)

MSE = np.mean(
    (y - y_pred) ** 2
)

SST = np.sum(
    (y - np.mean(y)) ** 2
)

R2 = 1 - (SSE / SST)

print("SSE =", SSE)
print("MSE =", MSE)
print("R² =", R2)
```

SSE = 9385.915871932419
MSE = 23.943662938603108
R² = 0.6059482578894348

```
# TASK 12: Plot regression and residual graphs

# Regression graph
```

```
plt.scatter(X, y)
plt.plot(X, y_pred)

plt.xlabel("Horsepower")
plt.ylabel("MPG")
plt.title("Regression Line")

plt.show()

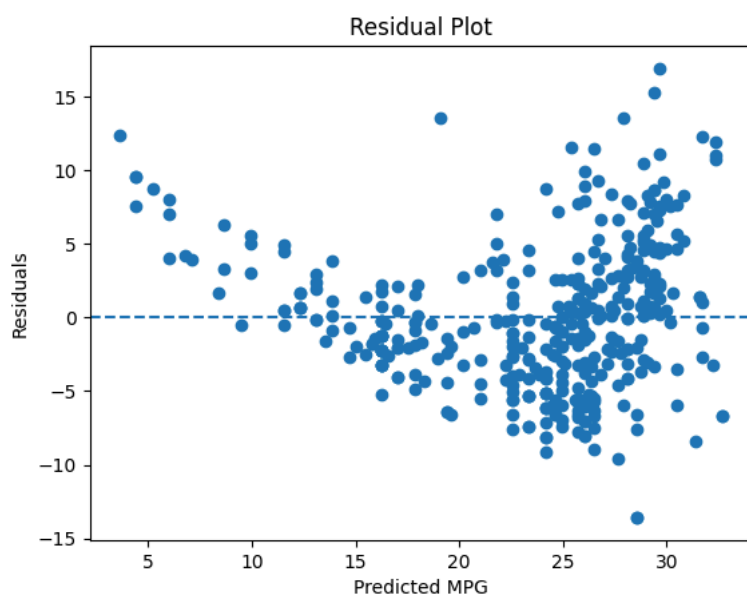
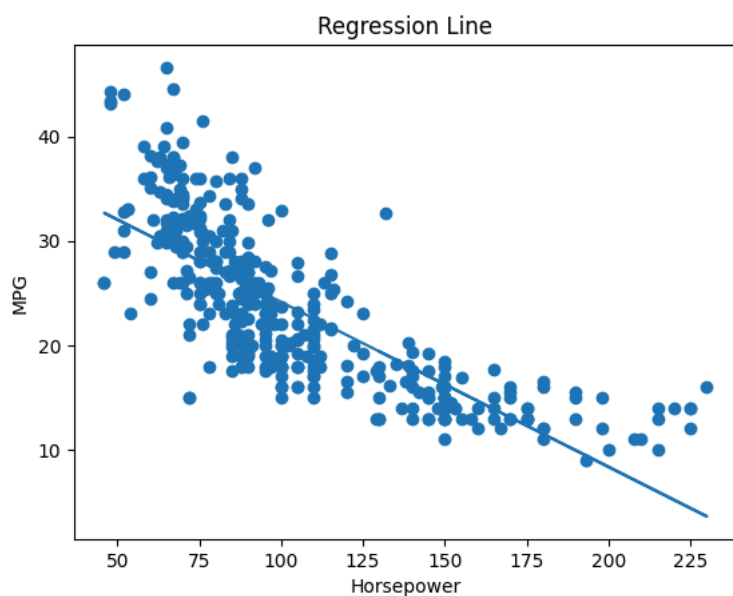
# Residual graph

plt.scatter(
    y_pred,
    residuals
)

plt.axhline(
    0,
    linestyle="--"
)

plt.xlabel("Predicted MPG")
plt.ylabel("Residuals")
plt.title("Residual Plot")

plt.show()
```



```
# TASK 13: Interpret results

print("Regression Equation:")

print(
    "MPG = {:.4f} + ({:.4f} × Horsepower)".format(
```

```

        w[0],
        w[1]
    )
)

if w[1] < 0:
    print(
        "Horsepower has a negative relationship with MPG."
    )
    print(
        "As horsepower increases, MPG decreases."
    )
else:
    print(
        "Horsepower has a positive relationship with MPG."
    )
    print(
        "As horsepower increases, MPG increases."
    )

print("R² =", R2)

```

Regression Equation:
 MPG = 39.9359 + (-0.1578 × Horsepower)
 Horsepower has a negative relationship with MPG.
 As horsepower increases, MPG decreases.
 R² = 0.6059482578894348

```
# TASK 14: Upload Google Colab onto GitHub repository
```

```

print(
    "Task 14: Upload the completed Google Colab notebook "
    "(.ipynb) to your GitHub repository."
)

```

Task 14: Upload the completed Google Colab notebook (.ipynb) to your GitHub repository.